Introduction

KnowFlow is a multi-tenant knowledge platform that turns uploaded documents into answerable, citable ground truth.

Every tenant owns its data outright; retrieval scopes every query to the requesting organization, never across it.

A document becomes useful only when it can be located, read, and cited, which is what this handbook describes.

Architecture

The ingest path is a small pipeline: upload stores the bytes, extraction turns them into pages, and chunking turns pages into retrieval units.

Pages keep the citation chain intact: each row records the page number, section anchor, and the s3:// source object.

Chunking never invents structure; it consumes the paragraphs and sections that extraction preserved and groups them with intent.

Components communicate over HTTP on the local stack: the backend owns uploads while the processor owns extraction and chunking.

Storage

Object storage holds the immutable originals; PostgreSQL holds pages and chunks; the vector index (later) holds embeddings.

Storage keys are tenant-scoped and uuid-suffixed so a filename collision can never cross a tenant boundary.

A chunk's metadata preserves the page range and section it came from, so a citation can point at both a page and a source.

Retrieval

Retrieval asks for chunks, then assembles an answer with citations back to the pages those chunks came from.

The atomic retrieval unit is the chunk: small enough to fit a context window, large enough to be self-contained.

Paragraph chunks are the production choice because they keep each retrieval unit semantically coherent.

Chunking

Fixed-size chunks give predictable byte prices but split sentences without regard to meaning.

Token-based chunks respect a model's context budget and are the standard baseline for RAG systems.

Overlapping windows share a tail between neighbours so context near a boundary is not lost.

Paragraph chunks group the author's own blocks, which makes the answer faithful to the document's structure.

A single enormous paragraph is the only adversary: it gets sliced at word boundaries, never merged with neighbours.

Security

Tenant separation is a schema invariant: every tenant-owned row carries organization_id and every query filters on it.

Row-level security policies will harden this later; the filter discipline is enforced by the application today.